UNICEF Education Monitoring Dashboard – Portfolio Case Study

Kano State, Nigeria

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1. Executive Summary

I developed a UNICEF-style Education Monitoring Dashboard to simulate a real-world education reporting system for Kano State, Nigeria.

This project involved generating a large synthetic dataset using Python, cleaning and validating the data in Microsoft Excel, and building an interactive dashboard in Power BI.

The dashboard enables stakeholders to monitor education performance, identify underserved communities, and support data-driven decision-making.

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2. Business Problem

Organizations such as UNICEF, NGOs, and government agencies require reliable education data to monitor key education indicators such as:

* Student enrollment
* Attendance rates
* Gender inclusion
* Infrastructure readiness
* WASH indicators

Without proper analytics systems, raw datasets become difficult to interpret, making decision-making slower and less effective.

This project addresses that challenge by transforming raw education monitoring data into interactive dashboards that provide actionable insights.

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3. Project Objective

The main goal of this project was to build a professional reporting system capable of converting education monitoring data into visual insights for decision-makers.

Key Objectives

* Monitor education performance across LGAs
* Identify low-performing communities
* Track gender disparities
* Evaluate WASH readiness
* Monitor trends over time
* Support evidence-based decision-making

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4. Data Collection Strategy

Since access to real UNICEF education data was unavailable, I designed and generated a realistic synthetic dataset using Python.

Dataset Characteristics

* **Total Records:** 7,500+
* **Coverage:** 12 LGAs in Kano State
* **Time Period:** 2024–2025
* **School Categories:** Multiple
* **Dataset Type:** Synthetic but realistic NGO-style monitoring data

This approach allowed simulation of real reporting conditions while preserving analytical flexibility.

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5. Technical Workflow

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Python → Excel → Power BI
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Python

Used to generate realistic synthetic education monitoring data.

Excel

Used for validation, cleaning, and structure review.

Power BI

Used for:

* Data modeling
* Relationship creation
* DAX calculations
* Interactive visualizations
* Dashboard design
* KPI reporting

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6. Dashboard Design

The dashboard contains five analytical pages.

Page 1 — Executive Summary

Provides high-level KPI overview:

* Total Enrollment
* Attendance Rate
* Girls Percentage
* Total Teachers
* Dropouts

Page 2 — LGA Performance Analysis

Compares education performance across Kano State LGAs to identify top-performing and underserved areas.

Page 3 — Gender & Inclusion

Analyzes:

* Girls enrollment
* Female teacher distribution
* Disability student inclusion
* Gender parity indicators

Page 4 — Facilities & WASH

Monitors:

* Water access

* Functional toilets
* Learning materials
* Infrastructure readiness

Page 5 — Performance Trends Over Time

Tracks monthly changes in:

* Enrollment
* Attendance
* Dropouts
* WASH indicators

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7. Key Insights

Insight 1 — Urban vs Rural Gap

Urban LGAs consistently demonstrated stronger attendance and enrollment performance compared with rural LGAs.

Insight 2 — Gender Inclusion

Girls participation improved in several LGAs, although rural communities still showed lower inclusion indicators.

Insight 3 — Infrastructure Challenges

Schools in rural communities experienced lower water access and reduced learning material availability.

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8. Challenges Encountered

Several challenges were encountered during project development:

* Building accurate data relationships
* Handling many-to-many relationship issues in Power BI
* Designing realistic synthetic datasets
* Creating optimized DAX measures
* Maintaining consistent dashboard layout and UX

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9. Solutions Implemented

The following solutions were implemented:

* Created lookup tables to improve relationships
* Optimized DAX calculations for KPI accuracy
* Used slicers and custom tooltips for interactivity
* Applied professional dashboard design principles
* Improved navigation with page buttons

These improvements enhanced usability, performance, and visual clarity.

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10. Business Impact

This dashboard demonstrates how education monitoring data can be transformed into actionable insights for:

* UNICEF reporting

* NGO program monitoring
* Government planning
* Donor reporting
* Resource allocation
* Policy decision-making

The dashboard improves visibility into performance gaps and helps prioritize intervention areas.

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11. Skills Demonstrated

Technical Skills

* Python
* Microsoft Excel
* Power BI
* DAX

Analytical Skills

* Data Cleaning
* Data Modeling
* Dashboard Design
* KPI Reporting
* Data Storytelling
* Business Analysis
* Problem Solving

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12. Final Outcome

This project reflects my ability to design end-to-end analytics solutions, from raw data generation to executive-level dashboard reporting.

It demonstrates practical capabilities relevant to roles in:

* NGOs

* UN Agencies

* Monitoring & Evaluation (M&E)

* Business Intelligence

* Data Analytics

The project showcases both technical and communication skills required to transform raw data into meaningful business insights.

Author

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